

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT Tool 3

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO3	Assessment:	Assessment Tool 3– Hackthon
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Assessment Tool 3– Hackthon		
Student Name:	T. Niranjan Kumar	Reg. No.:	192473007
Section / Batch:	CSE(DS)	Date:	31-08-2026

Code of Conduct

I **T. Niranjan Kumar (192473007)**(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: **T. Niranjan**

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Customer Segmentation for Marketing. Cluster mall/e-commerce customers based on annual income and spending score to identify target segments for a marketing campaign; use the Elbow Method and Silhouette Score to choose optimal k.

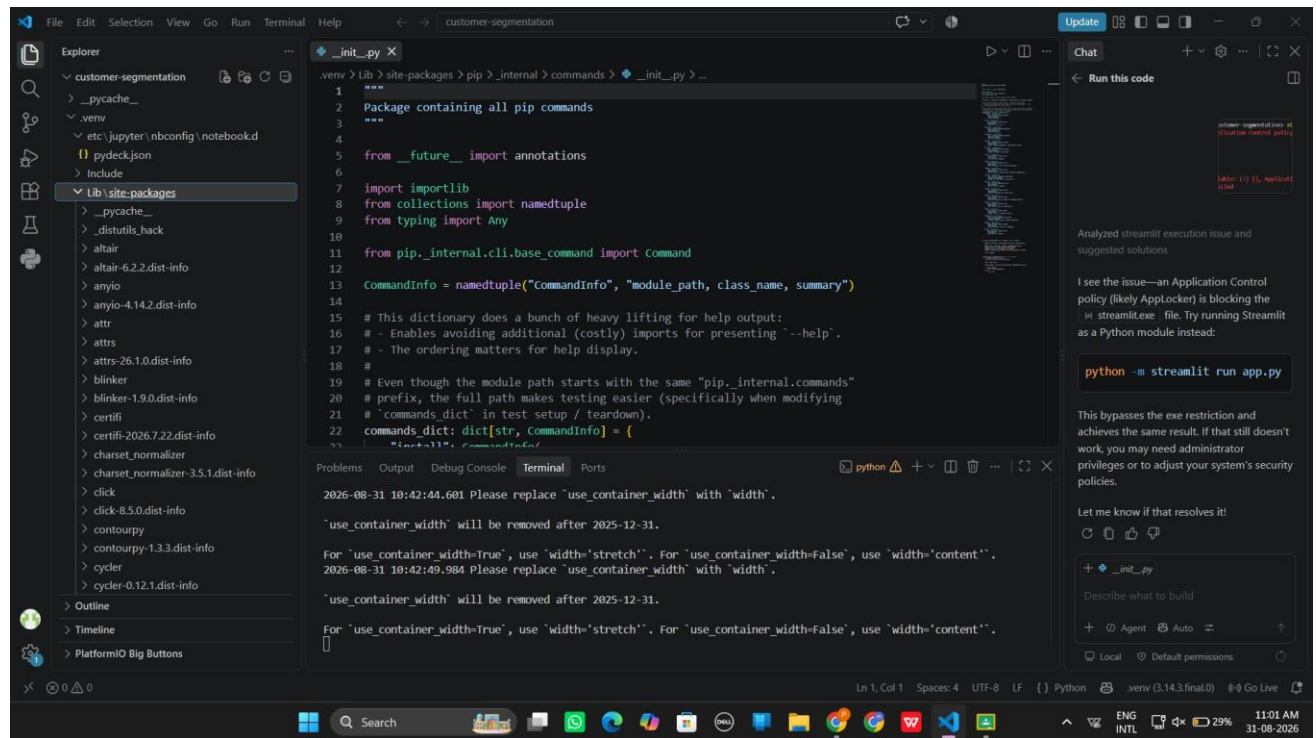
Problem Statement:

The objective of this project is to segment mall/e-commerce customers into distinct groups based on their annual income and spending score using clustering techniques. Customer segmentation helps businesses understand different customer behaviors and identify groups that can be targeted with customized marketing strategies.

The project will apply the K-Means clustering algorithm to group customers with similar income and spending patterns. The optimal number of clusters (k) will be determined using the Elbow Method and validated using the Silhouette Score. The resulting customer segments will then be analyzed to identify valuable target groups, such as high-income/high-spending customers, low-income/low-spending customers, and potential customers who could be targeted with specific promotional campaigns.

Goal: To identify meaningful customer segments that can help the business design more effective, personalized, and data-driven marketing campaigns.

CODE:

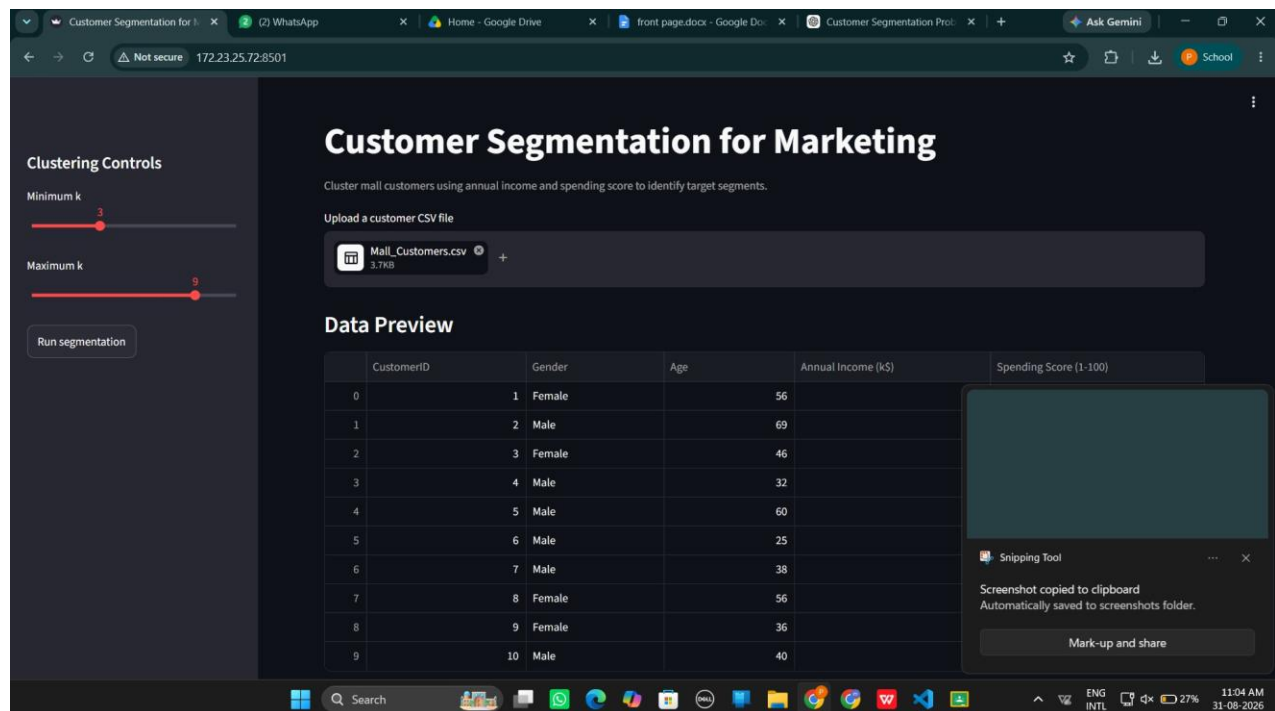


```
1 """
2 Package containing all pip commands
3 """
4
5 from __future__ import annotations
6
7 import importlib
8 from collections import namedtuple
9 from typing import Any
10
11 from pip._internal.cli.base_command import Command
12
13 CommandInfo = namedtuple("CommandInfo", "module_path, class_name, summary")
14
15 # This dictionary does a bunch of heavy lifting for help output:
16 # - Enables avoiding additional (costly) imports for presenting '--help'.
17 # - The ordering matters for help display.
18
19 # Even though the module path starts with the same "pip._internal.commands"
20 # prefix, the full path makes testing easier (specifically when modifying
21 # 'commands.dict' in test setup / teardown).
22 commands_dict: dict[str, CommandInfo] = {
23     "pip._internal.commands": CommandInfo
```

2026-08-31 10:42:44.601 Please replace 'use_container_width' with 'width'.
'use_container_width' will be removed after 2025-12-31.
For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content''.
2026-08-31 10:42:49.984 Please replace 'use_container_width' with 'width'.
'use_container_width' will be removed after 2025-12-31.
For 'use_container_width=True', use 'width='stretch''. For 'use_container_width=False', use 'width='content'.

python -m streamlit run app.py

OUTPUT:



Customer Segmentation for Marketing

Cluster mail customers using annual income and spending score to identify target segments.

Upload a customer CSV file

Mail_Customers.csv 1.7KB

Data Preview

	CustomerID	Gender	Age	Annual Income (k\$)	Spending Score (1-100)
0	1	Female		56	
1	2	Male		69	
2	3	Female		46	
3	4	Male		32	
4	5	Male		60	
5	6	Male		25	
6	7	Male		38	
7	8	Female		56	
8	9	Female		36	
9	10	Male		40	

Snipping Tool
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